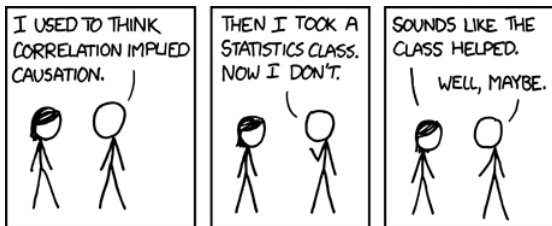


Advanced Applied Econometrics

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Thanks: slides heavily borrow from Scott Cunningham

What is regression discontinuity design?

Recently there has been an increase in a particular type of research design known as *regression discontinuity design* (RDD). Cook (2008) has a fascinating history of thought on how and why.

- Donald Campbell, educational psychologist, is the originator of regression discontinuity design (Thistlethwaite and Campbell, 1960).
- Lay dormant for decades (Cook 2008). Angrist and Lavy (1999) and Black (1999) appear to be the first papers in economics using it again.
- Pictures are helpful for understanding the RDD research design. Tell me what you think the following picture is saying.

RDD Visual Example

VOL. 1 NO. 1 CARPENTER AND DOBKIN: EFFECT OF ALCOHOL CONSUMPTION ON MORTALITY 177

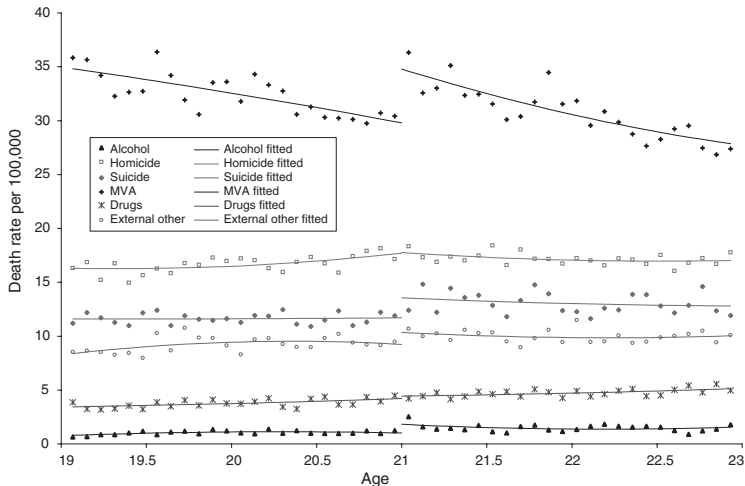
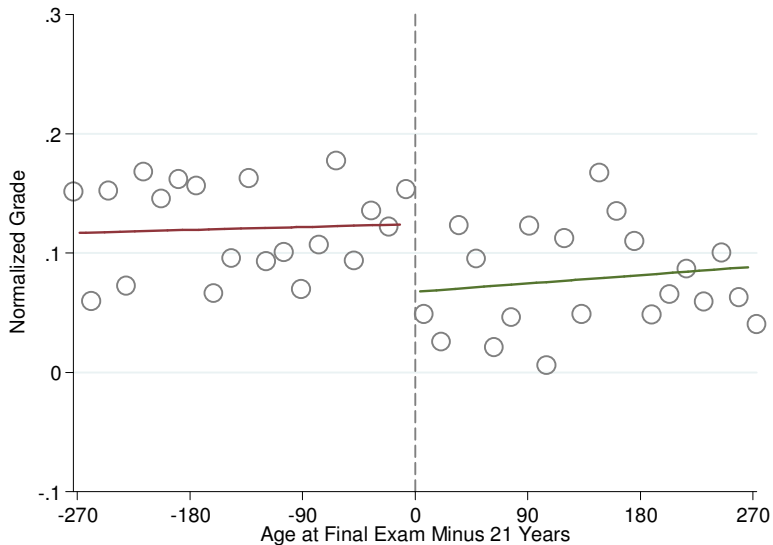


FIGURE 4. AGE PROFILES FOR DEATH RATES BY EXTERNAL CAUSE

Notes: See notes to Figure 3. The categories are mutually exclusive. The order of precedence is homicide, suicide, MVA, deaths with a mention of alcohol, and deaths with a mention of drugs. The ICD-9 and ICD-10 Codes are

RDD Visual Example

Figure 3: Regression Discontinuity Estimates of the Effect of Drinking on Achievement



What is a regression discontinuity design?

- RDD is based on a simple idea: the assignment of some treatment to observational units is completely known.
 - We know that the probability and amount of drinking alcohol “jumps” when a person becomes 21 (treatment)
 - Therefore we can use this knowledge to estimate the causal effect of some treatment (drinking) on some hypothesized outcome (drunk driving)
 - If the persons around age 21 are similar, then we have a *conditionally randomized experiment* .
- RDD is appropriate in situations where treatment assignment is sharply discontinuous in the values of a variable
- It's best understood with an example. We'll look at one of the first papers to use it in economics

RDD Data Requirements

- Question: So, where can I find these “jumps”? Answer: Humans are embedding “jumps” in their rules all the time. Dumb rules – while usually bad policy – are *great* for research.
- Validity doesn't require the assignment rule be arbitrary, only that it is known, precise and free of manipulation. The most effective RDD studies involve programs where X has a “hair trigger” that is not tightly related to the outcome being studied. Examples:
 - Probability of being tried as an adult (higher penalties for a given crime) “jumps” at age 18
 - Probability of being arrested for DWI “jumps” at blood alcohol content >0.08
 - Probability of receiving universal healthcare insurance “jumps” at age 65
 - Probability of receiving medical attention “jumps” when birthweight falls below 1,500 grams
 - Probability of having to attend summer school “jumps” when grade falls below 60
- Data requirements can be substantial. Large sample sizes are characteristic features of the RDD
 - If there are strong trends, one typically needs a lot of data
 - Researchers are typically using administrative data or settings such as birth records where there are **many** observations

RD's Relationship to IV

- RDD is really IV with control variables
- Redefine the measurement of R so that R is equal to zero at r_0
- Define $Z = \mathbf{1}(R \geq 0)$
- Then the coefficient on Z in a regression like

```
. reg Y Z R R2 R3
```

is the reduced form discontinuity, and

```
. reg X Z R R2 R3
```

is the first stage discontinuity

- The ratio of discontinuities is what is known as “fuzzy RDD” – this is an RD design where you are interested in the effect of X on Y
- A simple way to implement this is using IV

```
. ivregress 2sls Y (X=Z) R R2 R3
```

RD's Relationship to IV: Fear of Division

- The distinction between “fuzzy” and “sharp” RDD is akin to discussion surrounding “what is the right parameter” in the context of a social experiment
- Often, we have an experiment that is hard to characterize what exactly the treatment was. Consider the Moving to Opportunity Demonstration Project
 - Moving to Opportunity Demonstration Project randomly assigned households to new neighborhoods using randomized housing vouchers. But, not everyone complied. And it was a complex policy making it hard to know the mechanism by which the experiment may have improved outcomes.
 - This is sometimes known as “fear of division”, i.e., one might be more comfortable with presenting just the intention-to-treat (ITT) parameter which is just the reduced form regression of Y on Z , than with an IV-style parameter
- In RDD settings, a similar kind of conversation occurs to the one you will hear about in IV applications
- Many papers will not present an IV-style parameter, but rather a blizzard of ITT parameters, out of a “fear” of the exclusion restrictions may not hold

Sharp vs. Fuzzy RDD

RDD estimates the causal effect by distinguishing the discontinuous function, $1(X \geq c_0)$, from the smooth selection function, $f(X)$ (though Van der Klaauw uses S instead of X notation)

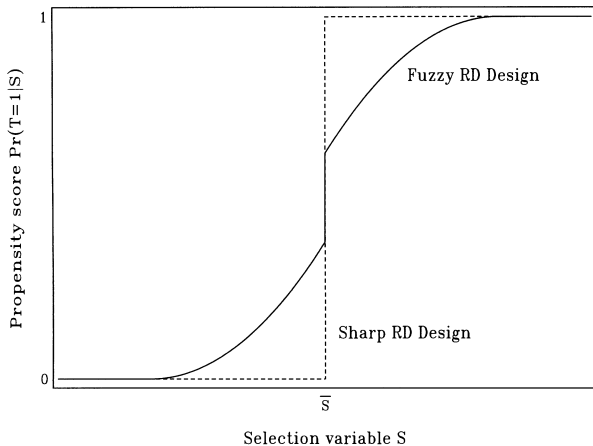


FIGURE 2

ASSIGNMENT IN THE SHARP (DASHED) AND FUZZY (SOLID) RD DESIGN

Sharp vs. Fuzzy RDD

- 1 Sharp RDD: Treatment is a deterministic function of running variable, X . Example: Medicare benefits.
- 2 Fuzzy RDD: Discontinuous “jump” in the *probability* of treatment when $X > c_0$. Cutoff is used as an instrumental variable for treatment. Example: Maimonides Rule (Angrist and Lavy 1999)

Treatment assignment in the sharp RDD

Deterministic treatment assignment (“sharp RDD”)

In Sharp RDD, treatment status is a deterministic and discontinuous function of a covariate, X_i :

$$D_i = \begin{cases} 1 & \text{if } X_i \geq c_0 \\ 0 & \text{if } X_i < c_0 \end{cases}$$

where c_0 is a known threshold or cutoff. In other words, if you know the value of X_i for a unit i , you know treatment assignment for unit i with certainty.

Example: Let X be age. Americans aged 64 are not eligible for Medicare, but Americans aged 65 ($X \geq c_{65}$) are eligible for Medicare (ignoring disability exemptions)

Implication: Sharp RDD relies on extrapolation across covariate values for its causal inference because there is no value of X_i (e.g., age) for which you observe both treatment and control observations. Therefore, we cannot be agnostic about regression functional form.

Treatment effect: definition and estimation

Definition of treatment effect

Assume constant treatment effects potential outcomes model linear in X :

$$Y_i^0 = \alpha + \beta X_i$$

$$Y_i^1 = Y_i^0 + \delta$$

Use the switching equation and write in terms of Y_i :

$$Y_i = Y_i^0 + (Y_i^1 - Y_i^0)D_i$$

$$Y_i = \alpha + \beta X_i + \delta D_i + \varepsilon_i$$

The treatment effect parameter, δ , is the discontinuity in the conditional expectation function:

$$\begin{aligned}\delta &= \lim_{X_i \rightarrow c_0} E[Y_i^1 | X_i = c_0] - \lim_{c_0 \leftarrow X_i} E[Y_i^0 | X_i = c_0] \\ &= \lim_{X_i \rightarrow c_0} E[Y_i | X_i = c_0] - \lim_{c_0 \leftarrow X_i} E[Y_i | X_i = c_0]\end{aligned}$$

The sharp RDD estimation is interpreted as an average causal effect of the treatment at the discontinuity

$$\delta_{SRD} = E[Y_i^1 - Y_i^0 | X_i = c_0]$$

Notice the role of *extrapolation* in estimating treatment effects when sharp RDD

- Left of cutoff, only non-treated observations, $D_i = 0$ for $X < c_0$
- Right of cutoff, only treated observations, $D_i = 1$ for $X \geq c_0$

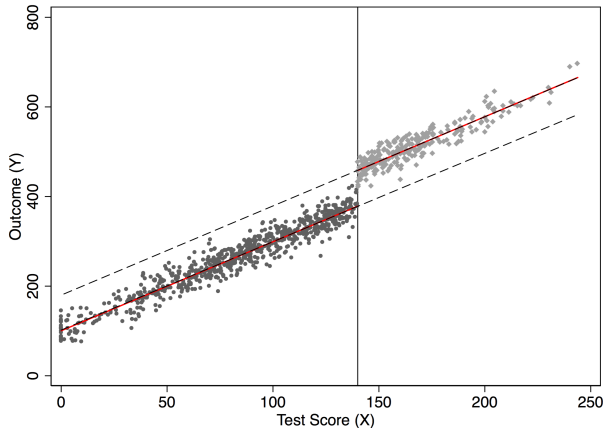


Figure: Dashed lines are extrapolations

Extrapolation and Continuous

- Notice the role of *extrapolation* in estimating treatment effects when sharp RDD
 - Left of cutoff, only non-treated observations, $D_i = 0$ for $X < c_0$
 - Right of cutoff, only treated observations, $D_i = 1$ for $X \geq c_0$
- The extrapolation is to a counterfactual. Recall the Rubin model:
 - Y_i^1 denotes outcome for person i if treated, Y_i^0 denotes *same* person i outcome if not treated
 - Causal effect for person i is $\delta_i = Y_i^1 - Y_i^0$ and the average treatment effect is $E[\delta_i] = E[Y_i^1 - Y_i^0]$
- In RDD, the counterfactuals are conditional on X . We are interested in the treatment effect at $X = c_0$:

$$E[Y_i^1 - Y_i^0 | X_i = c_0]$$

Key identifying assumption

Continuity of conditional regression functions (Hahn, Todd and Van der Klaauw 2001; Lee 2008)

$E[Y_i^0|X = c_0]$ and $E[Y_i^1|X = c_0]$ are continuous (smooth) in X at c_0 .

Remark Alfred Marshall quotes Darwin in Principles of Economics (1890), “*Natura non facit saltum*”, which means “nature does not make jumps”. Jumps are unnatural, so when they occur, they require an explanation

Meaning If population average *potential outcomes*, Y^1 and Y^0 , are continuous functions of X at the cutoff, c_0 , then potential average outcomes *do not* jump at c_0 . All other unobserved determinants of Y are continuously related to the running variable, X

Implication This assumption allows us to use average outcome of units right below the cutoff as a valid counterfactual for units right above the cutoff. The causal effect of the treatment will be based on extrapolation from the trend, $E[Y_i^0|X < c_0]$, to those values of $X > c_0$ for the $E[Y_i^0|X > c_0]$.

Conditional Independence

CIA

$Y^0, Y^1 \perp\!\!\!\perp D|X$. Once I fix on X , the running variable, D is independent of potential outcomes

- Trivial assumption (for the first time)
- There is no variation in the treatment variable, D , conditional on the running variable.
 - Once I condition on the running variable, there is no longer any variation left in treatment
- No common support – RDD is a local extrapolation outside the support of the data to predict mean treated and untreated potential outcomes at c_0


```
/// --- Examples using simulated data
set obs 1000
set seed 1234567

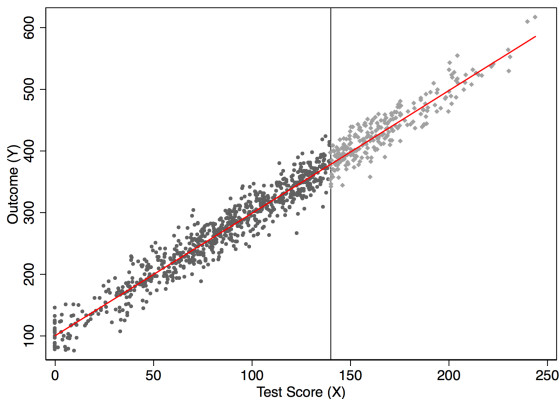
* Generate running variable
gen x = rnormal(100, 50)
replace x=0 if x < 0
drop if x > 280
sum x, det

* Set the cutoff at X=140. Treated if X > 140
gen D = 0
replace D = 1 if x > 140
```

Graphical example of continuous assumption

```
gen y1 = 100 + 0*D + 2*x + rnormal(0, 20)
```

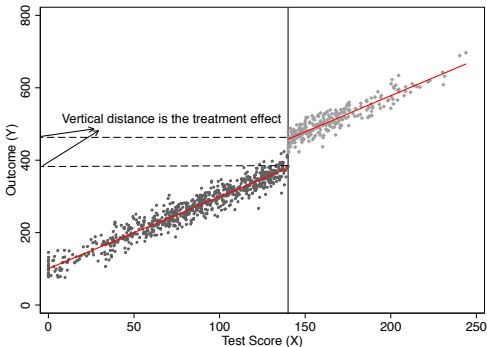
```
scatter y1 x if D==0, msize(vsmall) || scatter y1 x if D==1, msize(vsmall) le  
xline(140, lstyle(foreground)) || lfit y1 x if D ==0, color(red) || lfit y1  
x if D ==1, color(red) ytitle("Outcome (Y)") xtitle("Test Score (X)")
```



Graphical example with treatment effect, cutoff=140

```
gen y = 100 + 80*D + 2*x + rnormal(0, 20)
```

```
scatter y x if D==0, msize(vsmall) || scatter y x if D==1, ///  
msize(vsmall) legend(off) xline(140, lstyle(foreground)) || /// lfit y x if  
D ==0, color(red) || lfit y x if D ==1, color(red) /// ytitle("Outcome (Y)")  
xtitle("Test Score (X)")
```



Tangent: centering at the cutoff point

- It is common for authors to transform X by “centering” at c_0 :

$$Y_i = \alpha + \beta(X_i - c_0) + \delta D_i + \varepsilon_i$$

This doesn't change the interpretation of the treatment effect – only the interpretation of the intercept.

- Example: Medicare and age 65. Center the running variable (age) by subtracting 65:

$$\begin{aligned} Y &= \beta_0 + \beta_1(\text{Age} - 65) + \beta_2 \text{Edu} \\ &= \beta_0 + \beta_1 \text{Age} - \beta_1 65 + \beta_2 \text{Edu} \\ &= \alpha + \beta_1 \text{Age} + \beta_2 \text{Edu} \end{aligned}$$

where $\alpha = \beta_0 - \beta_1 65$. All other coefficients, notice, have the same interpretation, except for the intercept.

```
. gen x_c=x-140
```

```
. reg y D x
```

Source	SS	df	MS	Number of obs =	999
Model	15842893.9	2	7921446.97	F(2, 996) =	19988.47
Residual	394715.557	996	396.30076	Prob > F =	0.0000
				R-squared =	0.9757
				Adj R-squared =	0.9756
Total	16237609.5	998	16270.1498	Root MSE =	19.907

y	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
D	80.01418	2.144779	37.31	0.000	75.80537	84.22298
x	1.986975	.0186779	106.38	0.000	1.950322	2.023627
_cons	100.3885	1.70944	58.73	0.000	97.03397	103.743

```
. reg y D x_c
```

Source	SS	df	MS	Number of obs =	999
Model	15842893.9	2	7921446.97	F(2, 996) =	19988.47
Residual	394715.554	996	396.300757	Prob > F =	0.0000
				R-squared =	0.9757
				Adj R-squared =	0.9756
Total	16237609.5	998	16270.1498	Root MSE =	19.907

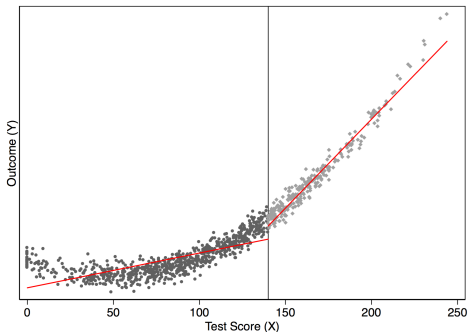
y	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
D	80.01418	2.144779	37.31	0.000	75.80537	84.22298
x_c	1.986975	.0186779	106.38	0.000	1.950322	2.023627
_cons	378.565	1.290755	293.29	0.000	376.032	381.0979

Nonlinearity bias

Smoothness in $E[Y_i^0|X_i]$ and linearity are different things. What if the trend relation $E[Y_i^0|X_i]$ does not jump at c_0 but rather is simply nonlinear? False positive problem.

```
gen x2 = x*x  
gen x3 = x*x*x  
gen y = 10000 + 0*D - 100*x +x2 + rnormal(0, 1000)
```

```
scatter y x if D==0, msize(vsmall) || scatter y x if D==1, msize(vsmall) legend(off) xline(140, lstyle(f  
ylabel(none) || lfit y x if D ==0, color(red) || lfit y x if D ==1, color(red) xtitle("Test Score (X)")  
yttitle("Outcome (Y)")
```



Sharp RDD: Nonlinear Case

- Suppose the nonlinear relationship is $E[Y_i^0|X_i] = f(X_i)$ for some reasonably smooth function $f(X_i)$. In that case we'd fit the regression model:

$$Y_i = f(X_i) + \delta D_i + \eta_i$$

- Since $f(X_i)$ is counterfactual for values of $X_i > c_0$, how will we model the nonlinearity? There are 2 ways of approximating $f(X_i)$:
 - ① Let $f(X_i)$ equal a p^{th} order polynomial:

$$Y_i = \alpha + \beta_1 x_i + \beta_2 x_i^2 + \cdots + \beta_p x_i^p + \delta D_i + \eta_i$$

- ② Use a nonparametric kernel method (later)

Different polynomials on the 2 sides of the discontinuity

- We can generalize the function, $f(x_i)$, by allowing the x_i terms to differ on both sides of the cutoff by including them both individually and interacting them with D_i . In that case we have:

$$\begin{aligned}E[Y_i^0|X_i] &= \alpha + \beta_{01}\tilde{X}_i + \beta_{02}\tilde{X}_i^2 + \cdots + \beta_{0p}\tilde{X}_i^p \\E[Y_i^1|X_i] &= \alpha + \delta + \beta_{11}\tilde{X}_i + \beta_{12}\tilde{X}_i^2 + \cdots + \beta_{1p}\tilde{X}_i^p\end{aligned}$$

where $\tilde{X}_i$ is the centered running variable (i.e., $X_i - c_0$). Centering at c_0 ensures that the treatment effect at $X_i = c_0$ is the coefficient on D_i in a regression model with interaction terms

- As Lee and Lemieux (2010) note, allowing different functions on both sides of the discontinuity should be the main results in an RDD paper as otherwise we use values from both sides of the cutoff to estimation the function on each side

Different polynomials on the 2 sides of the discontinuity

- To derive a regression model, first note that the observed values must be used in place of the potential outcomes:

$$E[Y|X] = E[Y^0|X] + (E[Y^1|X] - E[Y^0|X]) D$$

- Regression model you estimate is:

$$\begin{aligned} Y_i = & \alpha + \beta_{01}\tilde{x}_i + \beta_{02}\tilde{x}_i^2 + \cdots + \beta_{0p}\tilde{x}_i^p \\ & + \delta D_i + \beta_1^* D_i \tilde{x}_i + \beta_2^* D_i \tilde{x}_i^2 + \cdots + \beta_p^* D_i \tilde{x}_i^p + \varepsilon_i \end{aligned}$$

where $\beta_1^* = \beta_{11} - \beta_{01}$, $\beta_2^* = \beta_{21} - \beta_{02}$ and $\beta_p^* = \beta_{p1} - \beta_{0p}$

- The equation we looked at earlier a few slides back was just a special case of the above equation with $\beta_1^* = \beta_2^* = \beta_p^* = 0$
- The treatment effect at c_0 is δ
- The treatment effect at $X_i - c_0 = c > 0$ is: $\delta + \beta_1^* c + \beta_2^* c^2 + \cdots + \beta_p^* c^p$

Polynomial simulation example

```
capture drop y x2 x3
gen x2 = x*x
gen x3 = x*x*x
gen y = 10000 + 0*D - 100*x +x2 + rnormal(0, 1000)

reg y D x x2 x3

predict yhat

scatter y x if D==0, msize(vsmall) || scatter y x if D==1, msize(vsmall) legend(off)
xline(140, lstyle(foreground)) ylabel(none) || line yhat x if D ==0, color(red)
sort || line yhat x if D ==1, sort color(red) xtitle("Test Score (X)") ytitle("Outcome (Y)")
```

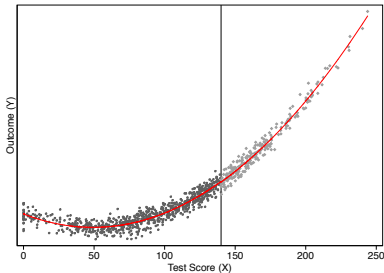


Figure: Third degree polynomial. Actual model second degree polynomial.

```
. gen x_c = x - 140
```

```
. gen x2_c = x2-140
```

```
. gen x3_c = x3-140
```

```
. reg y D x x2
```

Source	SS	df	MS	Number of obs =	999
Model	3.7863e+10	3	1.2621e+10	F(3, 995) =	13115.22
Residual	957507024	995	962318.617	Prob > F =	0.0000
				R-squared =	0.9753
				Adj R-squared =	0.9753
Total	3.8821e+10	998	38898361.8	Root MSE =	980.98

y	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
D	-115.5381	127.4967	-0.91	0.365	-365.7314	134.6552
x	-98.57582	2.285769	-43.13	0.000	-103.0613	-94.09034
x2	1.000001	.0122767	81.45	0.000	.9759098	1.024092
_cons	9864.218	111.1206	88.77	0.000	9646.16	10082.28

```
. reg y D x_c x2_c
```

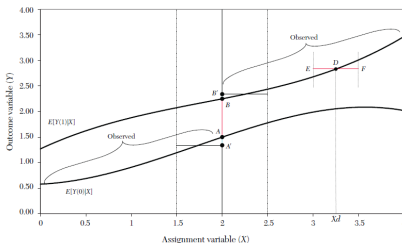
Source	SS	df	MS	Number of obs =	999
Model	3.7863e+10	3	1.2621e+10	F(3, 995) =	13115.22
Residual	957507020	995	962318.613	Prob > F =	0.0000
				R-squared =	0.9753
				Adj R-squared =	0.9753
Total	3.8821e+10	998	38898361.8	Root MSE =	980.98

y	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
D	-115.5381	127.4967	-0.91	0.365	-365.7315	134.6552
x_c	-98.57582	2.285769	-43.13	0.000	-103.0613	-94.09034
x2_c	1.000001	.0122767	81.45	0.000	.9759098	1.024092
_cons	-3796.397	227.7894	-16.67	0.000	-4243.4	-3349.394

Figure: P-order polynomial regressions with and without centering. Only intercept changes.

Kernel regression

- In addition to using p^{th} order polynomials to model the nonlinearities, we can use kernel regression. Neither is right or wrong – they have advantages and disadvantages.
- The nonparametric kernel method has problems because you are trying to estimate regressions at the cutoff point which results in a “boundary problem” (Hahn, Todd and Van der Klaauw 2001)



- While the “true” effect is AB , with a certain bandwidth a rectangular kernel would estimate the effect as $A'B'$
- There is therefore systematic bias with the kernel method if the $f(X)$ is upwards or downwards sloping

Kernel Method - Local linear regression

- The standard solution to this problem is to run local linear nonparametric regression (Hahn, Todd and Van der Klaauw 2001) – substantially reduces the bias
- Think of it as a weighted regression restricted to a window – kernel provides the weights to that regression.

$$(\hat{a}, \hat{b}) \equiv_{a,b} \sum_{i=1}^n (y_i - a - b(x_i - c_0))^2 K\left(\frac{x_i - c_0}{h}\right) 1(x_i > c_0)$$

where x_i is the value of the running variable, c_0 is the cutoff, K is a kernel function and $h > 0$ is a suitable bandwidth

Optimal bandwidths

- A rectangular kernel would give the same result as taking $E[Y]$ at a given bin on X . The triangular kernel gives more importance to the observations closer to the center.
- While estimating this in a given window of width h around the cutoff is straightforward, it's more difficult to choose this bandwidth (or window), and the method is sensitive to the choice of bandwidth. There is essentially a tradeoff between bias and efficiency. See Lee and Lemieux (2010) for two methods to choose on the bandwidth. This is an active area of research in econometrics.
- Imbens and Kalyanaraman (2012) propose a method for estimating optimal bandwidths which may differ on either side of the cutoff. This can be implemented now with `rdrobust`

Probabilistic treatment assignment (i.e. “fuzzy RDD”)

The probability of receiving treatment changes discontinuously at the cutoff, c_0 , but need not go from 0 to 1

$$\lim_{X_i \rightarrow c_0} \Pr(D_i = 1 | X_i = c_0) \neq \lim_{c_0 \leftarrow X_i} \Pr(D_i = 1 | X_i = c_0)$$

Examples: Incentives to participate in some program may change discontinuously at the cutoff but are not powerful enough to move everyone from non participation to participation.

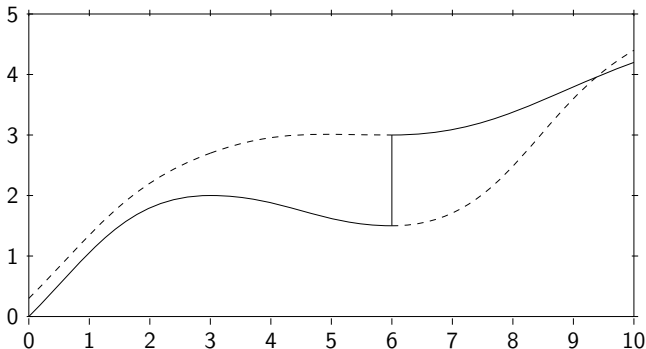
- In the sharp RDD, D_i was *determined* by $X_i \geq c_0$; in the fuzzy RDD, the conditional probability of treatment *jumps* at c_0 .
- The relationship between the probability of treatment and X_i can be written as:

$$P[D_i = 1 | X_i] = g_0(X_i) + [g_1(X_i) - g_0(X_i)] T_i$$

where $T_i = 1$ if $(X_i \geq c_0)$ and 0 otherwise.

Visualization of identification strategy (i.e. smoothness)

- Conditional expectation of the 2 potential outcomes given X for $D = 0, 1$
 - Partly dashed, partly solid $\rightarrow$ continuous functions of the covariate
- Conditional expectation of the observed outcome, Y
 - Solid $\rightarrow$ jumps at $X = 6$



Use the discontinuity as IV

Wald estimator of treatment effect under Fuzzy RDD

Average causal effect of the treatment is the Wald IV parameter

$$\delta_{\text{Fuzzy RDD}} = \frac{\lim_{X \rightarrow c_0} E[Y|X = c_0] - \lim_{c_0 \leftarrow X} E[Y|X = c_0]}{\lim_{X \rightarrow c_0} E[D|X = c_0] - \lim_{c_0 \leftarrow X} E[D|X = c_0]}$$

- Fuzzy RDD is numerically equivalent and conceptually similar to instrumental variables
 - Numerator: “jump” in the regression of the outcome on the running variable, X . “Reduced form”.
 - Denominator: “jump” in the regression of the treatment indicator on the running variable X . “First stage”.
- Use software package to estimate (e.g., `ivregress 2sls` in STATA).
- Same IV assumptions, caveats about compliers vs. defiers, and statistical tests that we discussed with instrumental variables apply here – e.g., check for weak instruments using F test on instrument in first stage, etc.

First stage relationship between X and D

- One can use both T_i as well as the interaction terms as instruments for D_i . If one uses only T_i as IV, then the it is a “just identified” model which usually has good finite sample properties.
- In the just identified case, the first stage would be:

$$D_i = \gamma_0 + \gamma_1 X_i + \gamma_2 X_i^2 + \cdots + \gamma_p X_i^p + \pi T_i + \zeta_{1i}$$

where π is the causal effect of T on the conditional probability of treatment.

- The fuzzy RD reduced form is:

$$Y_i = \mu + \kappa_1 X_i + \kappa_2 X_i^2 + \cdots + \kappa_p X_i^p + \rho \pi T_i + \zeta_{2i}$$

Fuzzy RDD with varying Treatment Effects - Second Stage

- As in the sharp RDD case one can allow the smooth function to be different on both sides of the discontinuity.
- The second stage model with interaction terms would be the same as before:

$$Y_i = \alpha + \beta_{01}\tilde{x}_i + \beta_{02}\tilde{x}_i^2 + \cdots + \beta_{0p}\tilde{x}_i^p \\ + \rho D_i + \beta_1^* D_i \tilde{x}_i + \beta_2^* D_i \tilde{x}_i^2 + \cdots + \beta_p^* D_i \tilde{x}_i^p + \eta_i$$

- Where $\tilde{x}$ are now not only normalized with respect to c_0 but are also fitted values obtained from the first stage regression.

Fuzzy RDD with Varying Treatment Effects - First Stages

- Again one can use both T_i as well as the interaction terms as instruments for D_i
- Only using T the estimated first stages would be:

$$\begin{aligned} D_i = & \gamma_{00} + \gamma_{01}\tilde{X}_i + \gamma_{02}\tilde{X}_i^2 + \cdots + \gamma_{0p}\tilde{X}_i^p \\ & + \pi T_i + \gamma_1^*\tilde{X}_i T_i + \gamma_2^*\tilde{X}_i^2 T_i + \cdots + \gamma_p^* T_i + \zeta_{1i} \end{aligned}$$

- We would also construct analogous first stages for $\tilde{X}_i D_i$, $\tilde{X}_i^2 D_i$, $\dots$, $\tilde{X}_i^p D_i$.

What does Fuzzy RDD Estimate?

- As Hahn, Todd and van der Klaauw (2001) point out, one needs the same assumptions as in the standard IV framework
- As with other binary IVs, the fuzzy RDD is estimating LATE: the average treatment effect for the compliers
- In RDD, the compliers are those whose treatment status changed as we moved the value of x_i from just to the left of c_0 to just to the right of c_0

Challenges to RDD

- Treatment is not as good as randomly assigned around the cutoff, c_0 , when agents are able to manipulate their running variable scores. This happens when:
 - 1 the assignment rule is known in advance
 - 2 agents are interested in adjusting
 - 3 agents have time to adjust
 - 4 Examples: re-take an exam, self-reported income, etc.
- Some other unobservable characteristic changes at the threshold, and this has a direct effect on the outcome.
 - In other words, the cutoff is endogenous
 - Example: Age thresholds used for policy (i.e., person turns 18, and faces more severe penalties for crime) is correlated with other variables that affect the outcome (i.e., graduation, voting rights, etc.)

Econometricians and applied social scientists have developed several formalized tests to evaluate the severity of these problems which we now discuss.

Test 1: Manipulation of the running variable

Sorting on the running variable (i.e., Manipulation)

Assume a desirable treatment, D , and an assignment rule $X \geq c_0$. If individuals sort into D by choosing X such that $X \geq c_0$, then we say individuals are sorting on the running variable.

- Motivating example: Suppose a doctor plans to randomly assign heart patients to a statin and a placebo to study the effect of the statin on heart attacks within 10 years. The doctor randomly assigns patients to two different waiting rooms, A and B , and plans to give those in A the statin and those in B the placebo. If some of the patients learn of the planned treatment assignment mechanism, what would we expect to happen? And how would you check for it?

McCrary Density Test

We would expect waiting room *A* to become *crowded*. In the RDD context, sorting on the running variable implies heaping on the “good side” of c_0

- McCrary (2008) suggests a formal test. Under the null the density should be continuous at the cutoff point. Under the alternative hypothesis, the density should increase at the kink (where D is viewed as good)
 - 1 Partition the assignment variable into bins and calculate frequencies (i.e., number of observations) in each bin
 - 2 Treat those frequency counts as dependent variable in a local linear regression
- The McCrary Density Test has become **mandatory** for every analysis using RDD.
 - If you can estimate the conditional expectations, you evidently have data on the running variable. So in principle you can always do a density test
 - You can download the (no longer supported) STATA ado package, DCdensity, to implement McCrary's density test (<http://eml.berkeley.edu/~jmccrary/DCdensity/>)
 - You can install rdd for R too (<http://cran.r-project.org/web/packages/rdd/rdd.pdf>)

Caveats about McCrary Density Test

- For RDD to be useful, you already need to know something about the mechanism generating the assignment variable and how susceptible it could be to manipulation. Note the rationality of economic actors that this test is built on.
- A discontinuity in the density is “suspicious” – it *suggests* manipulation of X around the cutoff is probably going on. In principle one doesn’t need continuity.
- This is a high-powered test. You need a lot of observations at c_0 to distinguish a discontinuity in the density from noise.

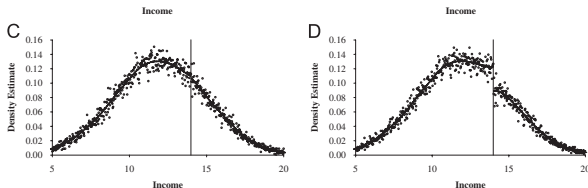
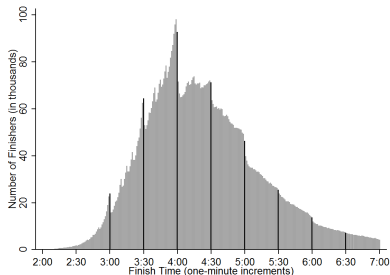


Figure: Panel C is density of income when there is no pre-announcement and no manipulation. Panel D is the density of income when there is pre-announcement and manipulation. From McCrary (2008).

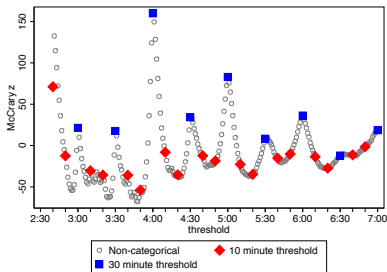
Visualizing manipulation

Figure 2: Distribution of marathon finishing times ($n = 9,378,546$)



NOTE: The dark bars highlight the density in the minute bin just prior to each 30 minute threshold.

Figure 3: Running McCrary z-statistic



NOTE: The McCrary test is run at each minute threshold from 2:40 to 7:00 to test whether there is a significant discontinuity in the density function at that threshold.

Figure: Figures 2 and 3 from Eric Allen, Patricia Dechow, Devin Pope and George Wu's (2013) "Reference-Dependent Preferences: Evidence from Marathon Runners".

http://faculty.chicagobooth.edu/devin.pope/research/pdf/Website_Marathons.pdf

More examples of testing for manipulation:

Lee (2008) Incumbency Effect

- David Lee's (2008) "Randomized Experiments from non-random selection in US House Elections", *Journal of Econometrics* analyzes the incumbency effect using Democratic incumbents for US congressional elections.
 - A large political science literature on the "incumbency advantage" – having won an election once helps you win subsequent elections.
 - Empirical challenge: how to separate incumbency advantage from selection – (i.e., candidates win multiple elections because they are better)
 - Identification: Incumbency is *assigned* to a candidate discontinuously at 50% voteshare under two-party democracy with majority rule.
- Lee (2008) analyzes the probability of winning the election in year $t + 1$ by comparing candidates who just won to candidates who just lost the election in year t .

Density test

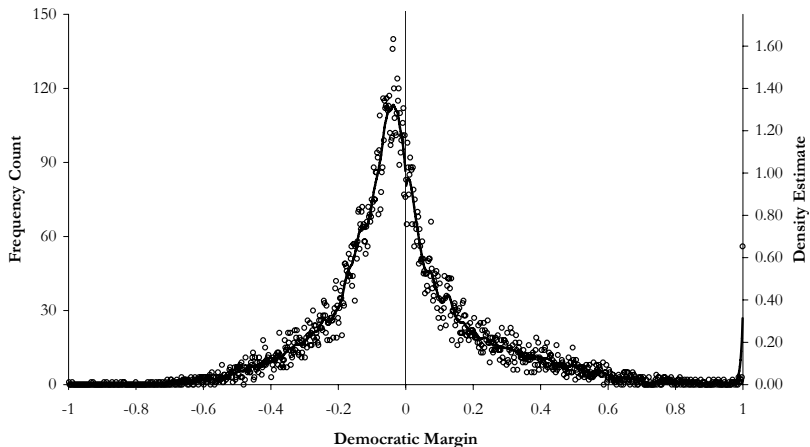


Figure: Democratic vote share relative to cutoff: popular elections to the House of Representatives, 1900-1990 (McCrary 2008).

More evidence of manipulation

Contrast this with roll call voting in the US House of Representatives

- Coordination is expected because these are repeated games, votes are public records, and side payments are possible in the form of future votes
- Bills around the cutoff are more likely to be passed than not. Seems like a good candidate for RDD
- Fails McCrary Density Test; **cannot** use RDD because policy decisions are not quasi-randomly assigned around cutoff

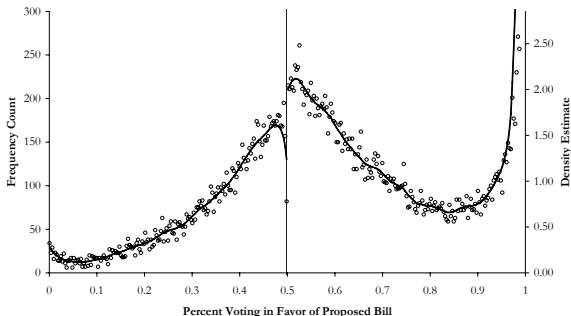


Figure: Percent voting Yeay: Roll Call Votes, US House of Representatives, 1857-2004 (McCrary 2008)

Test 2: Balance test on covariates

- This is a type of placebo test. For RDD to be valid in your study, you do not want to observe a discontinuity around the cutoff, c_0 , for average values of covariates that should **not** be affected by the treatment – e.g., pretreatment characteristics
- Question: What does a jump in the average values of pre-treatment characteristics have to do with the continuous (smoothness) assumption?
 - Choose other pre-treatment covariates, Z and do a similar graphical plot as you did for Y
 - You do **not** want to see a jump around the cutoff, c_0
 - A formal balance test involves the same procedure used to estimate the treatment effect, only use Z instead of Y as a LHS variable
 - Can combine test on multiple covariates into a single test statistic using seemingly unrelated regression (SUR) or a single “stacked” regression

Visualizing Placebo Test

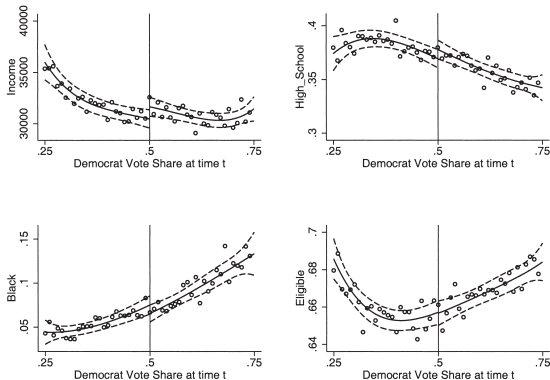


Figure: Figure 3 from Lee, Moretti and Butler (2004), “Do Voters Affect or Elect Policies?” *Quarterly Journal of Economics*. Panels refer to (top left to bottom right) the following district characteristics: real income, percentage with high-school degree, percentage black, percentage eligible to vote. Circles represent the average characteristic within intervals of 0.01 in Democratic vote share. The continuous line represents the predicted values from a fourth-order polynomial in vote share fitted separately for points above and below the 50 percent threshold. The dotted line represents the 95 percent confidence interval.

Test 3: Jumps at non-discontinuous points

- Imbens and Lemieux (2008) suggest to look at one side of the discontinuity (e.g., $X < c_0$), take the median value of the running variable in that section, and pretend it was a discontinuity, c'_0
- Then test whether in reality there is a discontinuity at c'_0 . You do **not** want to find anything.
- Another kind of placebo test. Similar to synthetic control falsification exercises where you look for non-effects in non-treatment periods. (See Abadie, Diamond and Hainmueller 2014)

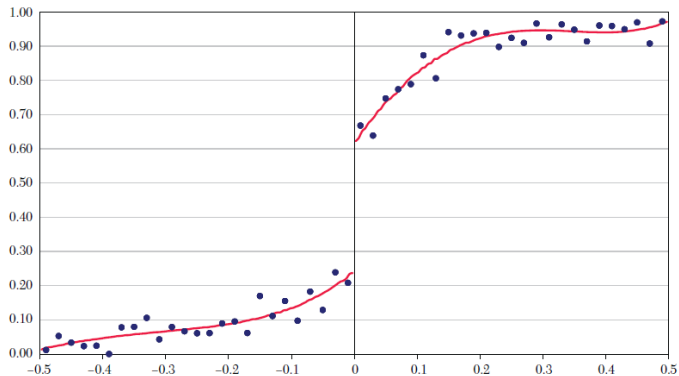
RDD graphs

A graphical analysis should be an integral part of any RDD.

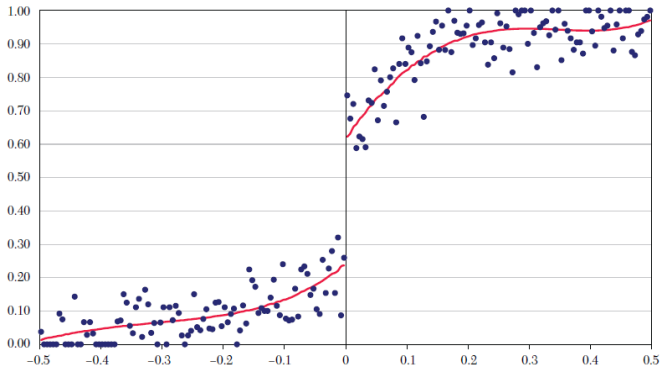
① Outcome by running variable, (X_i):

- The standard graph showing the discontinuity in the outcome variable
- Construct bins and average the outcome within bins on both sides of the cutoff
- You should also look at different bin sizes when constructing these graphs (see Lee and Lemieux, 2010, for details)
- Plot the running variables, X_i , on the horizontal axis and the average of Y_i for each bin on the vertical axis
- You may also want to plot a relatively flexible regression line on top of the bin means
- Inspect whether there is a discontinuity at c_0
- Inspect whether there are other unexpected discontinuities

Example: Outcomes by Running Variables



Example: Outcomes by Running Variables with smaller bins



More RDD Graphs

② Probability of treatment by running variable if fuzzy RDD

- In a fuzzy RDD, you also want to see that the treatment variable jumps at c_0
- This tells you whether you have a first stage

③ Covariates by a running variable

- Construct a similar graph to the one before but using a covariate as the “outcome”
- There should be **no jump** in other covariates at the discontinuity, c_0 .
- If the covariates jump at the discontinuity, one would doubt the identifying assumption

Example: Covariates by Running Variable

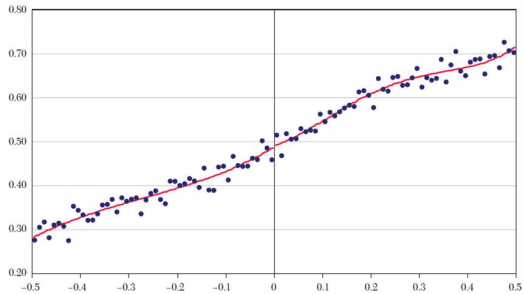


Figure 17. Discontinuity in Baseline Covariate (Share of Vote in Prior Election)

More RDD graphs!

4 The density of the running variable

- One should plot the number of observations in each bin.
- This plot allows to investigate whether there is a discontinuity in the distribution of the running variable at the threshold
- This would suggest that people can manipulate the running variable around the threshold.
- This is an indirect test of the identifying assumption that each individual has imprecise control over the assignment variable

Density of the running variable

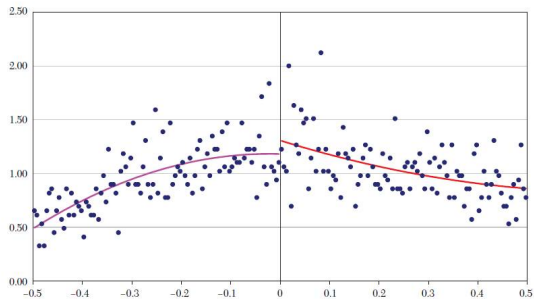


Figure 16. Density of the Forcing Variable (Vote Share in Previous Election)

“Do Voters Affect or Elect Policies?”

by Lee, Moretti and Butler (2004)

How do voters affect policy? There are two fundamentally different views of the role of elections in a representative democracy.

- ① **Convergence:** Heterogenous voter ideology forces each candidates to moderate their positions (e.g., median voter theorem).

“Competition for votes can force even the most partisan Republicans and Democrats to moderate their policy choices. In the extreme case, competition may be so strong that it leads to ‘full policy convergence’: opposing parties are forced to adopt identical policies” (Lee, Moretti, and Butler 2004; Downs 1957).

- ② **Divergence:** When partisan politicians cannot credibly commit to certain policies, then convergence is undermined. The result can be full policy *divergence*. Divergence is when the winning candidate, after taking office, simply pursues his most-preferred policy. In this case, voters fail to compel candidates to reach any kind of policy compromise.

Simplified model

- Candidates:** R and D are candidates in a Congressional race. Policy space is unidimensional where D 's and R 's per-period policy preferences are quadratic loss functions, $u(I)$ and $v(I)$ and I is the policy variable

D 's bliss point: $I^* = c(> 0)$

$$\begin{aligned}\max_I u(I) &= -\frac{1}{2}(I - c)^2 \\ \frac{\partial u(I)}{\partial I} &= 2 \times -\frac{1}{2}(I - c) = 0 \\ I^* &= c(> 0)\end{aligned}$$

R 's bliss point: $I^* = 0$

$$\begin{aligned}\max_I v(I) &= -\frac{1}{2}I^2 \\ \frac{\partial v(I)}{\partial I} &= -\frac{1}{2} \times 2I = 0 \\ I^* &= 0\end{aligned}$$

- Voters:** *Ex ante*, voters expect D (R) to choose policy x^e (y^e), and expect D to win with probability $P(x^e, y^e)$
 - When $x^e > y^e$ then $\frac{\partial P}{\partial x^e} > 0$, $\frac{\partial P}{\partial y^e} < 0$. Democrats could gain more votes by moderating the policy position.
 - P^* represents the underlying popularity of D party, or put differently, the probability D would win if $x = c$ and $y = 0$.
- if D (R) wins, policy x (y) will be implemented; rational expectations equilibrium assumed ($x = x^e$; $y = y^e$), game repeats in periods t and $t + 1$ where a period includes the election and the Congressional session.

Multiple Nash equilibria

① Partial/Complete Convergence: Voters affect policies

- Key result: $\frac{\partial x^*}{\partial P^*} > 0$ and $\frac{\partial y^*}{\partial P^*} > 0$
- If we did a “helicopter drop” of more Democrats in the district to exogenously increase P^* and this resulted in candidates changing their policy positions, then $\frac{\partial x^*}{\partial P^*} > 0$ and $\frac{\partial y^*}{\partial P^*} > 0$

② Complete divergence: Voters elect politicians with fixed policies

- Key result: $\frac{\partial x^*}{\partial P^*} = \frac{\partial y^*}{\partial P^*} = 0$
- An exogenous shock to P^* (e.g., the helicopter drop of Democrats) does *nothing* to equilibrium policies. Voters *elect* politicians' fixed policies.

Estimation

- Potential roll-call voting record outcomes of the representative in the district following election t is

$$RC_t = D_t x_t + (1 - D_t) y_t$$

where D_t is an indicator for whether D won the election. That is, only the winning candidate's policy is observed.

- This expression can be transformed into regression equations:

$$RC_t = \alpha_0 + \pi_0 P_t^* + \pi_1 D_t + \varepsilon_t \quad (1)$$

$$RC_{t+1} = \beta_0 + \pi_0 P_{t+1}^* + \pi_1 D_{t+1} + \varepsilon_{t+1} \quad (2)$$

where α_0 and β_0 are constants

- They parameterized the derivatives from the prior slide as π_0 , the coefficient on the probability, P . See their Appendix 1, p. 849 for details.
- This also allows an independent effect of the party, π_1

Estimation (cont.)

- Equation 198 cannot be directly estimated because we don't observe P^* . But suppose we could randomize D_t . Then D_t would be independent of P_t^* and ε_t . Then taking conditional expectations with respect to D_t we get:

$$\begin{aligned}
 \underbrace{E[RC_{t+1}|D_t = 1] - E[RC_{t+1}|D_t = 0]}_{\text{Observable}} &= \pi_0[P_{t+1}^{*D} - P_{t+1}^{*R}] \\
 &+ \underbrace{\pi_1[P_{t+1}^D - P_{t+1}^R]}_{\text{Observable}} \\
 &= \underbrace{\gamma}_{\text{Total effect of initial win on future roll call votes}} \quad (3)
 \end{aligned}$$

$$\underbrace{E[RC_t|D_t = 1] - E[RC_t|D_t = 0]}_{\text{Observable}} = \pi_1 \quad (4)$$

$$\underbrace{E[D_{t+1}|D_t = 1] - E[D_{t+1}|D_t = 0]}_{\text{Observable}} = P_{t+1}^D - P_{t+1}^R \quad (5)$$

Why this works

- The “elect” component is $\pi_1[P_{t+1}^D - P_{t+1}^R]$ and it's estimated as the difference in mean voting records between the parties at time t
- The fraction of districts won by Democrats in $t + 1$ is an estimate of $[P_{t+1}^D - P_{t+1}^R]$
- Because we can estimate the total effect, γ , of a Democrat victory in t on RC_{t+1} , we can net out the elect component to implicitly get the “affect” component
- Random assignment of D_t is crucial. Without it, equation 206 would reflect π_1 **and** selection (i.e., that Dem districts have more liberal bliss points).

Data and RDD Jargon

- Two main datasets: liberal voting score from the Americans for Democratic Action (ADA) linked with House of Representatives election results for 1946-1995
 - Authors use the ADA score for all US House Representatives from 1946 to 1995 as their voting record index
 - For each Congress, ADA chooses about twenty high-profile roll-call votes and creates an index varying 0 and 100 for each Representative of the House. Higher scores correspond to a more “liberal” voting record.
- RDD Jargon
 - The running variable is `voteshare` which is the share of all votes that went to a Democrat. ADA scores are linked to election returns data during that period.
 - They use exogenous variation in Democratic wins to check whether convergence or divergence is correct.
 - Discontinuity in the running variable occurs at `voteshare = 0.5`. When `voteshare > 0.5`, the Democratic candidate wins.
- Download STATA do file from the website: http://business.baylor.edu/scott_cunningham/teaching/causalinf/lmb2004.txt, save it on your computer, and rename the extension to say `lmb2004.do`. Open it in your STATA do-editor.

Statistical results

TABLE I
RESULTS BASED ON ADA SCORES—CLOSE ELECTIONS SAMPLE

Variable	Total effect			Elect component	Affect component
	γ	π_1	$(P_{t+1}^D - P_{t+1}^R)$	$\pi_1[(P_{t+1}^D - P_{t+1}^R)]$	$\pi_0[P_{t+1}^{*D} - P_{t+1}^{*R}]$
	ADA_{t+1}	ADA_t	DEM_{t+1}	(col. (2))*(col. (3))	(col. (1)) - (col. (4))
	(1)	(2)	(3)	(4)	(5)
Estimated gap	21.2 (1.9)	47.6 (1.3)	0.48 (0.02)	22.84 (2.2)	-1.64 (2.0)

Standard errors are in parentheses. The unit of observation is a district-congressional session. The sample includes only observations where the Democrat vote share at time t is strictly between 48 percent and 52 percent. The estimated gap is the difference in the average of the relevant variable for observations for which the Democrat vote share at time t is strictly between 50 percent and 52 percent and observations for which the Democrat vote share at time t is strictly between 48 percent and 50 percent. Time t and $t + 1$ refer to congressional sessions. ADA_t is the adjusted ADA voting score. Higher ADA scores correspond to more liberal roll-call voting records. Sample size is 915.

Figure: Lee, Moretti, and Butler 2004, Table 1.

```
. ** Tables - using OLS
. * Table 1: Results Based on ADA Scores
. reg score lagdemocrat if lagdemvoteshare>.48 & lagdemvoteshare<.52, cluster(id)
```

```
Linear regression                               Number of obs =      915
<snip>
```

		Robust				
score	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
lagdemocrat	21.28387	1.951234	10.91	0.000	17.45445	25.11329
_cons	31.19576	1.333983	23.39	0.000	28.57773	33.81378

```
. reg score democrat if lagdemvoteshare>.48 & lagdemvoteshare<.52, cluster(id)
```

```
Linear regression                               Number of obs =      915
<snip>
```

		Robust				
score	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
democrat	47.7056	1.356011	35.18	0.000	45.04434	50.36686
_cons	18.7469	.8432428	22.23	0.000	17.09198	20.40182

```
. reg democrat lagdemocrat if lagdemvoteshare>.48 & lagdemvoteshare<.52, cluster(id)
```

```
Linear regression                               Number of obs =      915
<snip>
```

		Robust				
democrat	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
lagdemocrat	.4843287	.0289322	16.74	0.000	.4275475	.5411099
_cons	.2417582	.0200939	12.03	0.000	.2023227	.2811938

Bandwidth, bias and efficiency

Relaxing bandwidth requirement is about trading off bias and efficiency

. * Table 1 (continued): Examine variations in model specification

. reg score democrat , cluster(id2)

Linear regression

Number of obs = 13588

<snip>

(Std. Err. adjusted for 505 clusters in id2)

score		Coef.	Robust Std. Err.	t	P> t	[95% Conf. Interval]

democrat		40.76266	1.495659	27.25	0.000	37.82416 43.70115
_cons		17.5756	.8225887	21.37	0.000	15.95947 19.19172

. reg score democrat if lagdemvoteshare>.48 & lagdemvoteshare<.52, cluster(id2)

Linear regression

Number of obs = 915

<snip>

(Std. Err. adjusted for 250 clusters in id2)

score		Coef.	Robust Std. Err.	t	P> t	[95% Conf. Interval]	

democrat		47.7056	2.043658	23.34	0.000	43.68054	51.73066
_cons		18.7469	1.38312	13.55	0.000	16.02279	21.47101

Control for the running variable?

```
. * Center vote share at 0.5 to improve interpretation in interaction models
.gen      demvoteshare_c = demvoteshare - 0.5
(11 missing values generated)

. * Next, control for the (centered) running variable as a linear control variable.
. * This is the simplest RDD.
.reg      score democrat demvoteshare_c, cluster(id2)
```

Linear regression Number of obs = 13577
<snip>

(Std. Err. adjusted for 505 clusters in id2)

	score	Coef.	Robust Std. Err.	t	P> t	[95% Conf. Interval]	
democrat		58.50236	1.555847	37.60	0.000	55.44561	61.5591
demvoteshare_c		-48.93761	4.441693	-11.02	0.000	-57.66412	-40.21109
_cons		11.03413	.9602657	11.49	0.000	9.147518	12.92075

```
. * Modeling the linearity such that slopes can differ above vs. below discontinuity
.xi: reg score i.democrat*demvoteshare_c, cluster(id2)
i.democrat      _Idemocrat_0-1      (naturally coded; _Idemocrat_0 omitted)
i.demo`t*demv~c  _IdemXdemvo_#      (coded as above)
```

Linear regression Number of obs = 13577
<snip>

(Std. Err. adjusted for 505 clusters in id2)

	score	Coef.	Robust Std. Err.	t	P> t	[95% Conf. Interval]	
_Idemocrat_1		55.43136	1.448568	38.27	0.000	52.58538	58.27734
demvoteshare_c		-5.682785	5.939863	-0.96	0.339	-17.35273	5.987156
_IdemXdemvo_1		-55.15188	8.236231	-6.70	0.000	-71.33346	-38.97031
_cons		16.81598	.9050024	18.58	0.000	15.03794	18.59403

Other bandwidths

```

.* Use +/-0.1 from cutoff (i.e., .4 to .6)
.xi: reg score i.democrat*demvoteshare_c if demvoteshare>.40 & demvoteshare<.60, cluster(id2)
i.democrat      _Idemocrat_0-1      (naturally coded; _Idemocrat_0 omitted)
i.demo*t*demv_c  _IdemXdemvo_#      (coded as above)

```

```

Linear regression              Number of obs =   4632
<snip>
(Std. Err. adjusted for 428 clusters in id2)

```

	score	Coef.	Robust Std. Err.	t	P> t	[95% Conf. Interval]	
_Idemocrat_1		47.15915	1.810413	26.05	0.000	43.60072	50.71758
demvoteshare_c		9.421594	18.33431	0.51	0.608	-26.61514	45.45832
_IdemXdemvo_1		-9.127629	31.8722	-0.29	0.775	-71.77357	53.51831
_cons		17.22656	1.239458	13.90	0.000	14.79036	19.66276

```

%* Use +/- 0.05
.xi: reg score i.democrat*demvoteshare_c if demvoteshare>.45 & demvoteshare<.55, cluster(id2)
i.democrat      _Idemocrat_0-1      (naturally coded; _Idemocrat_0 omitted)
i.demo*t*demv_c  _IdemXdemvo_#      (coded as above)

```

```

%Linear regression              Number of obs =   2387
%<snip>
%(Std. Err. adjusted for 367 clusters in id2)

```

	score	Coef.	Robust Std. Err.	t	P> t	[95% Conf. Interval]	
_Idemocrat_1		46.77845	2.491464	18.78	0.000	41.87906	51.67783
demvoteshare_c		54.82604	50.12314	1.09	0.275	-43.73946	153.3915
_IdemXdemvo_1		-91.1152	81.05893	-1.12	0.262	-250.5149	68.28449
_cons		18.09713	1.693516	10.69	0.000	14.76689	21.42737

```

%* Use +/- 0.01
.xi: reg score i.democrat*demvoteshare_c if demvoteshare>.49 & demvoteshare<.51, cluster(id2)
i.democrat      _Idemocrat_0-1      (naturally coded; _Idemocrat_0 omitted)
i.demo*t*demv_c  _IdemXdemvo_#      (coded as above)

```

```

%Linear regression              Number of obs =   441
%<snip>
%(Std. Err. adjusted for 160 clusters in id2)

```

	score	Coef.	Robust Std. Err.	t	P> t	[95% Conf. Interval]	
_Idemocrat_1		45.27375	5.813497	7.79	0.000	33.79212	56.75539
demvoteshare_c		55.75797	716.0422	0.08	0.938	-1358.423	1469.939
_IdemXdemvo_1		-88.56438	955.1282	-0.09	0.926	-1974.939	1797.81
_cons		20.60166	4.514799	4.56	0.000	11.68495	29.51837

No voteshare control vs. polynomials

```
. * Narrow bandwidth, no vote share control (this is comparing means close to cutoff)
. reg score democrat if demvoteshare>.49 & demvoteshare<=.51, cluster(id2)
```

```
Linear regression      Number of obs =    441
                      F( 1,   159) = 244.90
                      Prob > F      = 0.0000
                      R-squared      = 0.5525
                      Root MSE    = 20.456
```

(Std. Err. adjusted for 160 clusters in id2)

	score	Coef.	Robust Std. Err.	t	P> t	[95% Conf. Interval]	
democrat		45.3931	2.900628	15.65	0.000	39.66438	51.12183
_cons		20.31812	2.026848	10.02	0.000	16.3151	24.32114

```
. * What about a polynomial of degree 5?
```

```
. gen demvoteshare2=demvoteshare^2
```

```
(11 missing values generated)
```

```
. gen demvoteshare3=demvoteshare^3
```

```
(11 missing values generated)
```

```
. gen demvoteshare4=demvoteshare^4
```

```
(11 missing values generated)
```

```
. gen demvoteshare5=demvoteshare^5
```

```
(11 missing values generated)
```

```
. reg score demvoteshare demvoteshare2 demvoteshare3 demvoteshare4 demvoteshare5 demvoteshare5 democrat, cluster(id2)
```

```
note: demvoteshare5 omitted because of collinearity
```

```
Linear regression      Number of obs =   13577
<snip>
```

(Std. Err. adjusted for 505 clusters in id2)

	score	Coef.	Robust Std. Err.	t	P> t	[95% Conf. Interval]	
demvoteshare		-19.30281	71.97002	-0.27	0.789	-160.701	122.0954
demvoteshare2		424.0805	543.3939	0.78	0.436	-643.5157	1491.677
demvoteshare3		-1746.451	1475.699	-1.18	0.237	-4645.731	1152.828
demvoteshare4		2559.367	1666.599	1.54	0.125	-714.9694	5833.704
demvoteshare5		-1248.399	664.662	-1.88	0.061	-2554.249	57.45029
demvoteshare5		0	(omitted)				
democrat		49.13039	1.764906	27.84	0.000	45.66291	52.59787
_cons		16.36452	3.07522	5.32	0.000	10.32269	22.40635

Polynomial with interactions

```

. * Center the running variable and use polynomials and interactions to model
* the nonlinearities below and above discontinuity
. gen x_c=demvoteshare-0.5
(11 missing values generated)

. gen x2_c=x_c^2
(11 missing values generated)

. gen x3_c=x_c^3
(11 missing values generated)

. gen x4_c=x_c^4
(11 missing values generated)

. gen x5_c=x_c^5
(11 missing values generated)

.
. reg score i.democrat##(c.x_c c.x2_c c.x3_c c.x4_c c.x5_c)

```

Source	SS	df	MS	Number of obs =	13577
Model	6677033.81	11	607003.073	F(11, 13565) =	1058.32
Residual	7780253.69	13565	573.553534	Prob > F =	0.0000
				R-squared =	0.4618
				Adj R-squared =	0.4614
Total	14457287.5	13576	1064.91511	Root MSE =	23.949

score	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]
1.democrat	47.73325	2.042906	23.37	0.000	43.72887 51.73763
x_c	-28.79765	74.09167	-0.39	0.698	-174.0276 116.4323
x2_c	-1138.232	1144.497	-0.99	0.320	-3381.605 1105.141
x3_c	-10681.29	7137.315	-1.50	0.135	-24671.42 3308.839
x4_c	-33490.23	18844.92	-1.78	0.076	-70428.88 3448.424
x5_c	-32873.77	17212.08	-1.91	0.056	-66611.83 864.302
democrat#c.x_c					
1	-5.793828	97.79088	-0.06	0.953	-197.4775 185.8899
democrat#c.x2_c					
1	1768.225	1433.252	1.23	0.217	-1041.149 4577.599
democrat#c.x3_c					
1	6279.346	8553.95	0.73	0.463	-10487.58 23046.28
democrat#c.x4_c					
1	47111.44	21834.51	2.16	0.031	4312.767 89910.11
democrat#c.x5_c					
1	17786.85	19488.24	0.91	0.361	-20412.81 55986.51
_cons	17.05847	1.469785	11.61	0.000	14.17749 19.93946

More polynomial and bandwidth regressions

```
. * We could limit the regressions to a window using a 2nd degree polynomial
. * interacted with the treatment variable
. reg score i.democrat##(c.x_c c.x2_c) if (demvoteshare>0.4 & demvoteshare<0.6)
```

Source	SS	df	MS	Number of obs = 4632		
				F(5, 4626) = 1153.29		
Model	2622762.02	5	524552.404	Prob > F = 0.0000		
Residual	2104043.2	4626	454.829918	R-squared = 0.5549		
				Adj R-squared = 0.5544		
Total	4726805.22	4631	1020.6878	Root MSE = 21.327		

score	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
1.democrat	45.9283	1.892566	24.27	0.000	42.21797	49.63863
x_c	38.63988	60.77525	0.64	0.525	-80.5086	157.7884
x2_c	295.1723	594.3159	0.50	0.619	-869.9704	1460.315
democrat#c.x_c						
1	6.507415	88.51418	0.07	0.941	-167.0226	180.0374
democrat#c.x2_c						
1	-744.0247	862.0435	-0.86	0.388	-2434.041	945.9916
_cons	17.71198	1.310861	13.51	0.000	15.14207	20.28189

Nonparametric estimation

- Hahn, Todd and Van der Klaauw (2001) clarified assumptions about RDD (i.e., continuity in conditional expectation regression functions)
- Also framed estimation as a non-parametric problem and emphasized using local polynomial regressions
- Nonparametric methods mean a lot of different things to different people in statistics.
- In RDD context, the idea is to estimate a model that doesn't assume a functional form for the relationship between Y (outcome variable) and X (running variable)
- That model would be something general like

$$Y = f(X) + \varepsilon$$

- A very basic method is to calculate $E[Y]$ for each bin on X – like a histogram.
- STATA has an option to do this called `cmogram` and it has a lot of useful options. We can recreate Figures I, IIA and IIB using it

Graphical estimate of equation 205

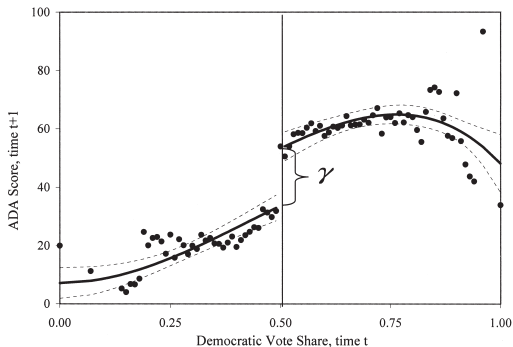


FIGURE I

Total Effect of Initial Win on Future ADA Scores: γ

This figure plots ADA scores after the election at time $t + 1$ against the Democrat vote share, time t . Each circle is the average ADA score within 0.01 intervals of the Democrat vote share. Solid lines are fitted values from fourth-order polynomial regressions on either side of the discontinuity. Dotted lines are pointwise 95 percent confidence intervals. The discontinuity gap estimates

$$\gamma = \underbrace{\pi_0(P_{t+1}^{*D} - P_{t+1}^{*R})}_{\text{"Affect"}} + \underbrace{\pi_1(P_{t+1}^{*D} - P_{t+1}^{*R})}_{\text{"Elect"}}.$$

Figure: Lee, Moretti, and Butler 2004, Figure I. $\gamma \approx 20$

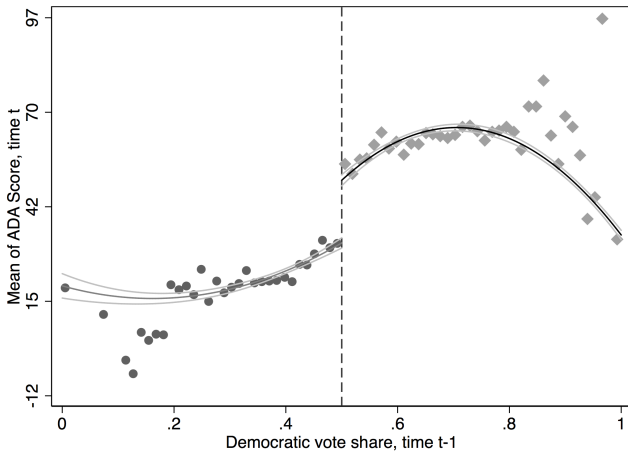
Nonparametric estimation with cmogran

```
. cmogran score lagdemvoteshare , cut(0.5) scatter line(0.5) qfitci
```

Plotting mean of score, conditional on lagdemvoteshare.

n = 13577

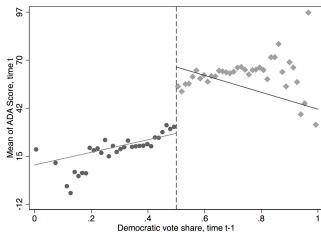
```
Bin #1: [0,.0135101361854656] (n = 181) (mean = 18.90983436977007)
Bin #2: (.0135101361854656,.0270202723709312] (n = 0) (mean = .)
Bin #3: (.0270202723709312,.0405304085563968] (n = 0) (mean = .)
Bin #4: (.0405304085563968,.0540405447418624] (n = 0) (mean = .)
Bin #5: (.0540405447418624,.067550680927328] (n = 0) (mean = .)
Bin #6: (.067550680927328,.0810608171127936] (n = 2) (mean = 11.2350001335144)
<snip>
Bin #36: (.9605263157894751,.9736842105263173] (n = 2) (mean = 96.58000183105469)
Bin #37: (.9736842105263173,.9868421052631594] (n = 0) (mean = .)
Bin #38: (.9868421052631594,1] (n = 1646) (mean = 33.014483642933)
```



Linear and lowess fits

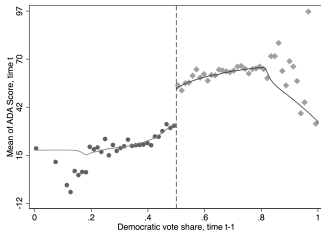
```
. cmogram score lagdemvoteshare , cut(0.5) scatter line(0.5) lfit
```

Plotting mean of score, conditional on lagdemvoteshare.



```
. cmogram score lagdemvoteshare , cut(0.5) scatter line(0.5) lowess
```

Plotting mean of score, conditional on lagdemvoteshare.



Graphical estimate of equation 206

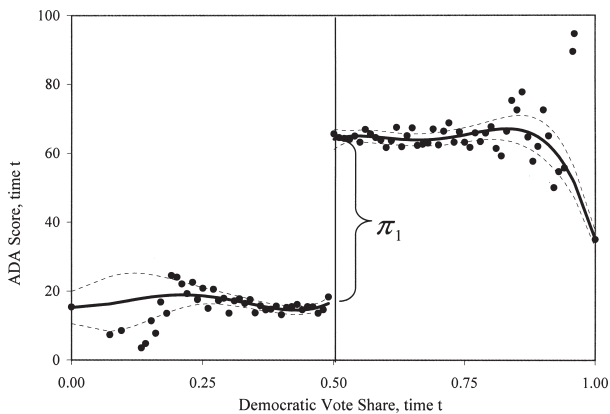


FIGURE IIa
Effect of Party Affiliation: π_1

Figure: Lee, Moretti, and Butler 2004, Figure IIa. $\pi_1 \approx 45$

* Lowess
clogram score demvoteshare, cut(.5) scatter line(.5) lowess

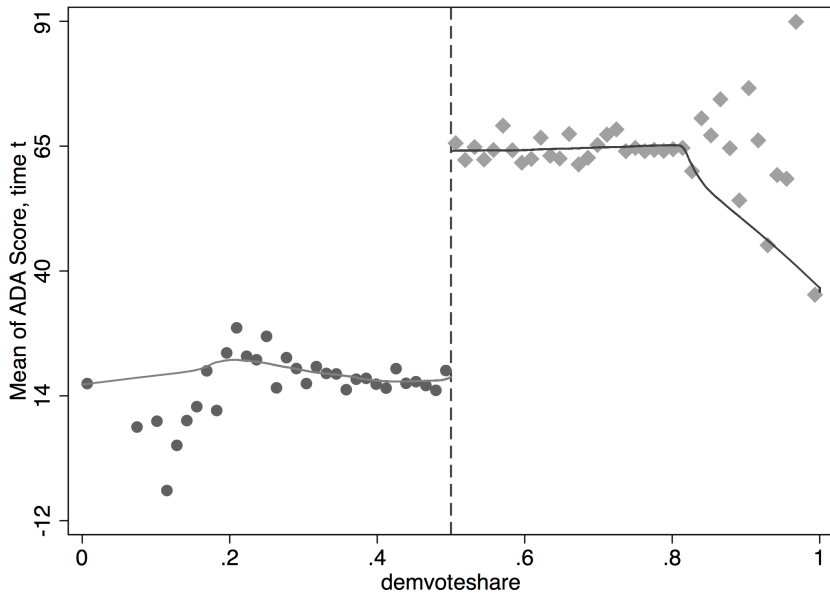


Figure: Cubic feet with confidence intervals

Graphical estimate of equation 207

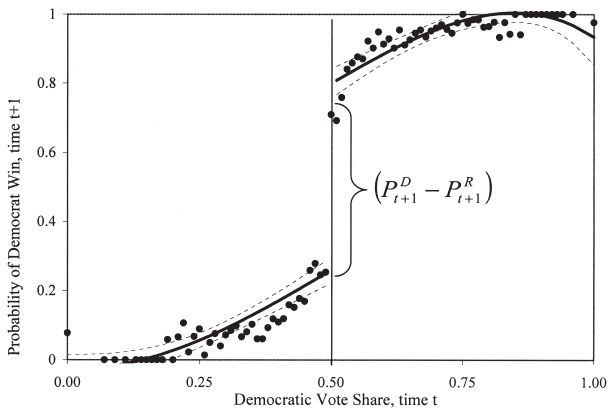


FIGURE IIb
Effect of Initial Win on Winning Next Election: $(P_{t+1}^D - P_{t+1}^R)$

Figure: Lee, Moretti, and Butler 2004, Figure IIb. $(P_{t+1}^D - P_{t+1}^R) \approx 0.50$

Kernel weighted local polynomial regression

- Hahn, Todd and Van der Klaauw (2001) showed that the one-sided kernel estimation (such as `lowess`) may have poor properties because the point of interest is at a boundary (i.e., the discontinuity), called the “boundary problem”
- They proposed to use “local linear nonparametric regressions” instead
- STATA’s `poly` estimates kernel-weighted local polynomial regressions. Think of it as a weighted regression restricted to a window like we’ve been doing (hence “local”) where the kernel provides the weights
- A rectangular kernel would give the same result as $E[Y]$ at a given bin on X . The triangular kernel gives more importance to observations close to the center.
- This method will be sensitive to how large the bandwidth (window) you choose

```

. * Note kernel-weighted local polynomial regression is a smoothing method.
. lpoly score demvoteshare if democrat == 0, nograph kernel(triangle) gen(x0 sdem0) ///
>      bwidth(0.1)

. lpoly score demvoteshare if democrat == 1, nograph kernel(triangle) gen(x1 sdem1) ///
>      bwidth(0.1)

. scatter sdem1 x1, color(red) msize(small) || scatter sdem0 x0, msize(small) color(red) ///
>      xline(0.5,lstyle(dot)) legend(off) xtitle("Democratic vote share") ytitle("ADA score")

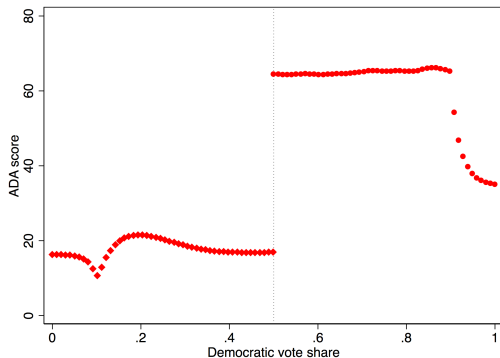
. * Next, let's get the treatment effect at the cutoff where demvoteshare=0.5
. capture drop sdem0 sdem1
. gen forat=0.5 in 1
. lpoly score demvoteshare if democrat==0, nograph kernel(triangle) gen(sdem0) at(forat) bwidth(0.1)
. lpoly score demvoteshare if democrat==1, nograph kernel(triangle) gen(sdem1) at(forat) bwidth(0.1)
. gen late=sdem1 - sdem0
. list sdem1 sdem0 late in 1/1

```

```

+-----+
|      sdem1      sdem0      late |
+-----+
1. | 64.395204  16.908821  47.48639 |
+-----+

```

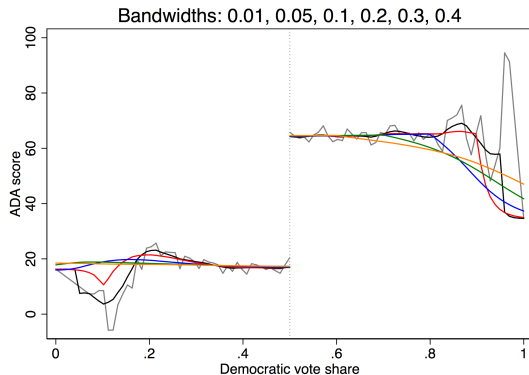


```

. * What happens when we change the bandwidth? Use 0.01, 0.05, 0.2, 0.3, 0.4
. capture drop smoothdem0* smoothdem1* x0* x1*
. local co 0
. foreach i in 0.01 0.05 0.1 0.20 0.30 0.40
2.   local co = 'co' +1
3.   lpoly score demvotesshare if democrat == 0, nograph kernel(triangle) gen(x0'co' smoothdem0'co') ///
>     bwidth('i')
4.   lpoly score demvotesshare if democrat == 1, nograph kernel(triangle) gen(x1'co' smoothdem1'co') ///
>     bwidth('i')
5.

. line smoothdem01 x01, msize(small) color(gray) sort || line smoothdem11 x11, sort color(gray) || ///
>   line smoothdem02 x02, color(black) sort || line smoothdem12 x12, sort color(black) || ///
>   line smoothdem03 x03, color(red) sort || line smoothdem13 x13, sort color(red) || ///
>   line smoothdem04 x04, color(blue) sort || line smoothdem14 x14, sort color(blue) || ///
>   line smoothdem05 x05, color(green) sort || line smoothdem15 x15, sort color(green) || ///
>   line smoothdem06 x06, color(orange) sort || line smoothdem16 x16, sort color(orange) ///
>   xline(0.5, lstyle(dot)) legend(off) xtitle("Democratic vote share") ytitle("ADA score") ///
>   title("Bandwidths: 0.01, 0.05, 0.1, 0.2, 0.3, 0.4")

```



- Several methods for choosing the optimal bandwidth (window), but it's always a trade off between bias and variance
- In practical applications, you want to check for balance around that window
- Standard error of the treatment effects can be bootstrapped but there are also other alternatives
- You could add other variables to nonparametric methods.
- Calonico, Cattaneo and Titiunik (2013b) propose local-polynomial regression discontinuity estimators with robust confidence intervals
- STATA ado package and R package are both called `rdrobust`

```
. rdrobust score demvoteshare, c(0.5) all bwselect(IK)
Preparing data.
Computing Bandwidth Selectors.
Computing Variance-Covariance Matrix.
Computing RD Estimates.
Estimation Completed.
```

Sharp RD Estimates using Local Polynomial Regression.

Cutoff c = .5 Left of c	Right of c	Number of obs =	13577	
-----+-----	-----	NN Matches =	3	
Number of obs	3535	3318	BW Type =	IK
Order Loc. Poly. (p)	1	1	Kernel Type =	Triangular
Order Bias (q)	2	2		
BW Loc. Poly. (h)	0.152	0.152		
BW Bias (b)	0.191	0.191		
rho (h/b)	0.795	0.795		

Outcome: score. Running Variable: demvoteshare.

Method	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]
-----+-----					
Conventional	47.171	.98058	48.1046	0.000	45.2488 49.0926
Robust	-	-	36.4839	0.000	44.0965 49.1034

All Estimates.

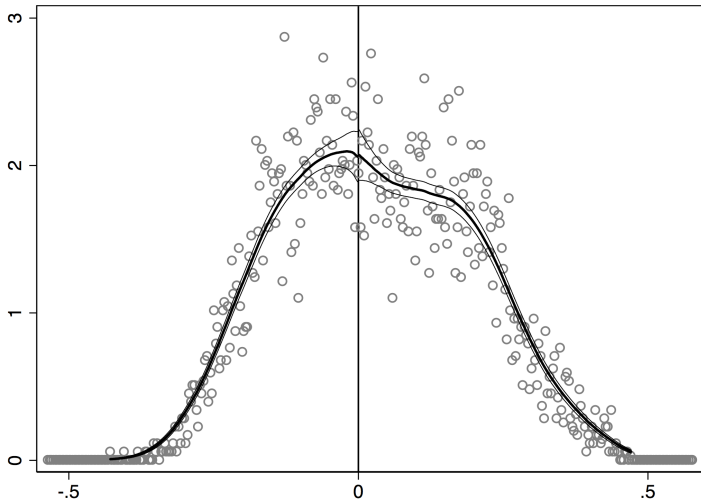
Method	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]
-----+-----					
Conventional	47.171	.98058	48.1046	0.000	45.2488 49.0926
Bias-Corrected	46.6	.98058	47.5226	0.000	44.678 48.5219
Robust	46.6	1.2773	36.4839	0.000	44.0965 49.1034

McCrary Density Test

```
. * McCrary (2008) density test to check for manipulation of the running variable (DCdensity)
. DCdensity demvoteshare_c if (demvoteshare_c>=0.5 & demvoteshare_c<0.5), breakpoint(0) generate(Xj Yj r0 fhat se_fhat)
Using default bin size calculation, bin size = .003047982
Using default bandwidth calculation, bandwidth = .104944836
```

```
Discontinuity estimate (log difference in height): .011195629
                                                    (.061618519)
```

```
Performing LLR smoothing.
296 iterations will be performed
```



“Pioneers take the arrows in the back”

– Jim West during a faculty meeting

Lee (2008) is important for the econometric theory, but this particular application turned out to be *wrong*:

“Contrary to the assumptions of RD, we show that bare winners and bare losers in US House elections (1942-2008) differ markedly on pretreatment covariates. Bare winners possess large ex ante financial, experience, and incumbency advantages over their opponents and are usually the candidates predicted to win by Congressional Quarterly’s pre-election ratings. Covariate imbalance actually worsens in the closest House election. National partisan tides help explain these patterns. We present evidence suggesting that sorting in close House elections is due mainly to activities on or before Election Day rather than post election recounts or other manipulation. The sorting is so strong that it is impossible to achieve covariate balance between matched treated and control observations, making covariate adjustment a dubious enterprise. Although RD is problematic for postwar House elections, this example does highlight the design’s advantages over alternatives: RD’s assumptions are clear and weaker than model-based alternatives, and their implications are empirically testable.” (Caughey and Sekhon, 2011).

Regression Kink Design

- Card, Lee, Pei and Weber (2012) introduces a variant of the RDD which they call regression kink design (RKD)
- They essentially use a kink in some policy rule to identify the causal effect of the policy
- Instead of a jump in the outcome you now expect a jump in the first derivative

Unemployment benefits in Austria

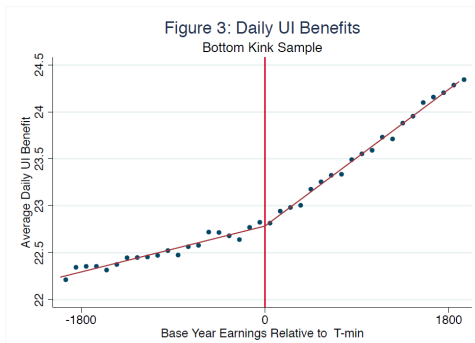
- They apply their design to answer the question whether the level of unemployment benefits affects the length of unemployment in Austria
- Unemployment benefits are based on income in a base period
- The benefit formula for unemployment exhibits 2 kinks
 - There is a minimum benefit level that isn't binding for people with low earnings
 - Then benefits are 55% of the earnings in the base period
 - There is a maximum benefit level that is adjusted every year
- People with dependents get small supplements (which is the reason why one can distinguish five “solid” lines in the following graph)
- Not everyone receives benefits that correspond one to one to the formula because of mistakes in the administrative data

Base Year Earnings and Unemployment Benefits



- The graph shows unemployment benefits (vertical axis) as a function of pre-unemployment earnings (horizontal axis)

Base Year Earnings and Benefits for Single Individuals



- Bin-size: 100 euros
- For single individuals UI benefits are flat below the cutoff. The relationship is still upward sloping because of family benefits.

Time to Next Job for Single Individuals



- People with higher base earnings have less trouble finding a job (negative slope)
- There is a kink: the relationship becomes shallower once benefits increase more.